Concept Source Table - Template

Scope: List all references that constitute the original sources for the VAST display:

- Isenberg, D. J. (1986). Group polarization: A critical review and meta-analysis. *Journal of Personality and Social Psychology*, *50*(6), 1141–1151. https://doi.org/10.1037/0022-3514.50.6.1141
- Burnstein, E., & Vinokur, A. (1977). Persuasive argumentation and social comparison as determinants of attitude polarization. *Journal of Experimental Social Psychology*, *13*(4), 315–332. https://doi.org/10.1016/0022-1031(77)90002-6
- Banisch, S., & Shamon, H. (2025). Biased processing and opinion polarization: Experimental refinement of argument communication theory in the context of the energy debate. Sociological Methods & Research, 54(1), 187–236. https://doi.org/10.1177/00491241231186658

Construct Source Table:

Reference	**Quote**	**Extracted element** (Concept or relationship)	**Element type** (**C**oncept **or** **R**elationship)	**Short name / associated concept**	**ID**	**im VAST (Y/N)**
Isenberg, D. J. (1986), p. 1141	Group polarization is said to occur when an initial tendency of individual group members toward a given direction is enhanced following group discussion Thus, on decisions in which group members have, on the average, a	Group polarization	Phenomenon	GP	**1**	**Y**
		Initial Tendency of group members with a moderate average proclivity	Concept	T_i^{pre}	2	Y

	moderate proclivity in a given direction, group discussion results in a more extreme average proclivity in the same direction.	Group Discussion	Higher-order concept	GD	3	Y
		Resulting Tendency of group members with a more extreme average proclivity	Concept	T_i^{post}	4	Y
Isenberg, D. J. (1986), p. 1145	PAT holds that an individual's choice or position on an issue is a function of the number and persuasiveness of pro and con arguments that that person recalls from memory when formulating his or her own position.	Individual's initial arguments that the person recalls from memory [Initial argument set]	Concept	IA_i	5	Y
		Number of arguments	Concept	n_i^{IA}	6	Y
		Persuasiveness of pro and con [direction] arguments	Concept	d_j, p_j	7	Y
		Individual's initial choice or position (tendency) on an issue is a function of the number and persuasiveness of pro and con arguments	Relationship		8	Y

		that person recalls from memory				
Isenberg, D. J. (1986), p. 1145	Group discussion will cause an individual to shift in a given direction to the extent that the discussion exposes that individual to persuasive arguments favoring that direction.	Additional arguments available following a group discussion	Concept	AA_i^{avail}	9	Y
		[number of additional arguments person i considers after group discussion]	Concept	n_i^{AA}	10	Y
		Individual (choice) shift	Higher-order concept	CS	11	Y
		(Choice) shift is caused by exposition to persuasive arguments in a group discussion	Relationship		12	Y
Isenberg, D. J. (1986), p. 1145	Two factors determine how persuasive a given argument will be. One factor is the perceived validity of the argument. How true is the argument? Does the argument fit into the person's previous views? Does the argument logically follow from	persuasiveness of a given argument	Concept	p_j	13	Y
		one factor of persuasiveness is perceived validity of argument	Concept	val_j	14	Y

	accepted facts or assumptions? The second factor determining persuasiveness is the perceived novelty of the argument. Does the argument represent a new way of organizing information? Does the argument suggest new ideas? Does the argument increase the perceiver's access to additional information that is stored in memory?	second factor of persuasiveness is perceived novelty of argument	Concept	nov_j	15	Y
		persuasiveness of other´s arguments consists of validity and novelty of arguments	relationship		16	Y
Developed independently for completeness	At the group level, there is an initial tendency with a moderate average proclivity, followed by a polarized tendency after the group discussion	Initial Tendency of group with a moderate average proclivity	Concept	T^{pre}	17	Y
		Tendency of group after group discussion	Concept	T^{post}	18	Y

Banisch, S., & Shamon, H. (2025), p. 191	Biased argument processing [...] refers to a person's tendency to inflate the quality of arguments that align with his or her existing attitude on an attitude object whereas the quality of those arguments that speak against a person's prevailing attitude are downgraded.	Congruence-Based Selective Uptake	Concept	SU	19	Y
Developed independently for completeness	Subset of additional arguments made available through group discussion that is selectively taken up and integrated into the individual's argument set as a function of prior tendency	selectively integrated additional arguments and beta	Concept	AA_i^{int}	20	Y
Developed independently for code alignment	In the simulation, the probability that a congruent additional argument is taken up increases with the strength parameter beta	Strength of congruence-based selective uptake	Concept	beta	21	N
Developed independently for completeness	Selectively integrated additional arguments are sampled from the pool of additional arguments	Selective uptake transforms AA_i^{avail} into AA_i^{int} conditional on	Relationship		22	Y

	available as a function of prior tendency and the strength of selective uptake (beta).	T_i^{pre} and beta				

Variable Table:

List all concepts that are not higher-order concepts (because higher-order concepts typically are not reducible to a single metric):

Construct	**Short Name**	**Scale level**	**Range/Values**	**Anchors**
Initial Tendency of group with a moderate average proclivity	T^{pre}	continuous	[-1;+1]	-1 = extremely negative opinion on a certain topic +1 = extremely affirmative opinion on a certain topic
Tendency of group after group discussion	T^{post}	continuous	[-1;+1]	-1 = extremely negative opinion on a certain topic +1 = extremely affirmative opinion on a certain topic
Initial Tendency of group members with a moderate average proclivity	T_i^{pre}	continuous	[-1;+1]	-1 = extremely negative opinion on a certain topic +1 = extremely affirmative opinion on a certain topic
Resulting Tendency of group members with a more extreme average proclivity	T_i^{post}	continuous	[-1;+1]	-1 = extremely negative opinion on a certain topic +1 = extremely affirmative opinion on a certain topic
Number of arguments	n_i^{AA}, n_i^{IA}	continuous / integer	0;1;2;… (discrete) $x \in \mathbb{N}0$	
persuasiveness	p_j	continuous	[0;1]	0 = argument not persuasive at all 1 = argument maximally convincing

direction	d_j	dichotomous	+1/-1	-1 = Contra Argument +1 = Pro Argument
Strength of congruence-based selective uptake	beta	ordinal	0;1;2	0 = no selective uptake / baseline 1 = moderate selective uptake 2 = strong selective uptake

Relationships Table:

Every variable in the VAST to which a one-sided arrow is pointing is a dependent (endogenous) variable. Provide a mathematical function for each endogenous variable, that takes all incoming arrows as input variables.

Output variable	**Function to compute the output variable**	**Comment**	**To what extent derived from theory?[a]**	**To what extent empirically backed?**
IA_i	$IA_i = (a_1, a_2, \ldots, a_n)$ mit $a_j = (d_j, p_j)$ fuer $j = 1, 2 \ldots, n$		Derived from theory	ID 8
AA_i^{avail}	$AA_i^{avail} = (a_{n+1}, a_{n+2}, \ldots, a_{n+m})$ $a_k = (d_k, p_k)$ fuer $k = n+1, n+2, \ldots, n+$	parallel to IA_i	Loosely inspired by theory	ID 9
T_i^{post}	$T_i^{post} = \frac{1}{n_i^{IA} + n_i^{AA}} \left(\sum_{j \in IA_i} p_j d_j + \sum_{j \in AA_i^{int}} p_j d_j \right)$		Loosely inspired by theory	ID 4

T_i^{pre}	$T_i^{\text{pre}} = \frac{1}{n} \sum_{j \in IA_i} p_j d_j$		Loosely inspired by theory	ID 2
T^{pre}	$\bar{T}^{\text{pre}} = \frac{1}{N} \sum_{i=1}^{N} T_i^{\text{pre}}$		Derived from theory	ID 17
T^{post}	$\bar{T}^{post} = \frac{1}{N} \sum_{i=1}^{N} T_i^{post}$		Derived from theory	ID 18
AA_i^{int}	$w_{ij} = \exp(\beta \cdot T_i^{pre} \cdot val_j)$ $val_j = p_j d_j$ $AA_i^{int} \sim \text{Sample}(AA_i^{avail}, n_i^{AA,int}, prob \propto w_{ij})$	beta = 0: zufällige Aufnahme beta > 0: kongruente Argumente werden wahrscheinlicher integriert	Loosely inspired by theory	ID 19; ID 20; ID 22

[a] Response scale:

- Dictated by theory (add IDs from the Construct Source Table as reference)
- Derived from theory

- Loosely inspired by theory
- Not based on focal theory, but rather on common sense or other theories
- Not at all.